Simulation-Based Inference for an Adaptive-Network SIR Model

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1 Introduction

[TODO: Model motivation, data description, why likelihood is intractable. Suggested length: ∼1 page.]

2 Model and Data

[TODO: Formal model description: Erdős–Rényi network ($N = 200$, $p = 0.05$, 5 initially infected), three synchronous operations (Infection, Recovery, Rewiring), and why the adaptive rewiring makes the likelihood intractable. Reference Gross et al. (2006). Suggested length: ∼0.5 pages.]

The three unknown parameters have independent uniform priors: $\beta \sim \mathcal{U}(0.05, 0.50)$, $\gamma \sim \mathcal{U}(0.02, 0.20)$, $\rho \sim \mathcal{U}(0.0, 0.8)$, with prior standard deviations $\sigma_\beta^{\text{prior}} = 0.130$, $\sigma_\gamma^{\text{prior}} = 0.052$, $\sigma_\rho^{\text{prior}} = 0.231$. The observed dataset comprises $R = 40$ independent replicates, each providing an infection time series, a rewiring time series, and a final degree distribution (Figure 1).

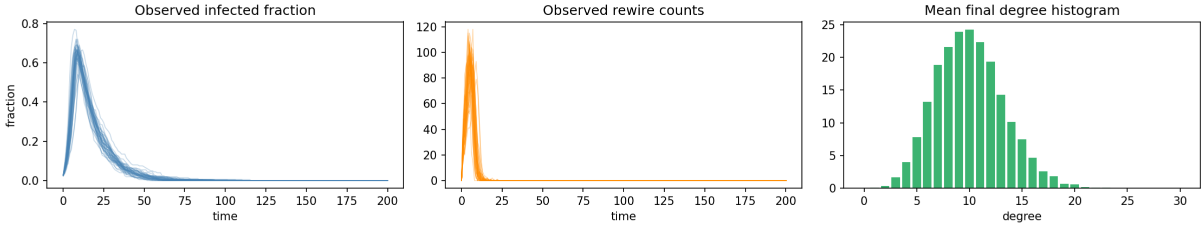

Figure 1: Observed data across 40 replicates: infection fraction (left), rewiring events (centre), final degree distribution (right).

3 Summary Statistics

Because acceptance probability in ABC falls exponentially with summary dimension (Blum et al., 2013), we compress the raw time series to a small set of scalar features. A naive approach is to concatenate all available scalars into one vector. The problem is that in a standardised Euclidean distance, a modality with more features contributes proportionally more to the total distance — a simulation can be accepted purely because its infection curve is close, even if its rewiring and degree statistics are far off. We therefore design a *balanced* set of 15 features with deliberate equal representation across the three modalities: 5 infection, 4 rewiring, and 6 degree summaries, all averaged across replicates. Within each group, features are chosen for identifiability rather than completeness (e.g. `epidemic_duration` over `final_inf` to better separate β from ρ; cross-replicate standard deviations to capture stochastic variability; degree quartiles and entropy over isolated/ high-degree fractions to describe the full distribution shape).

3.1 Identifiability diagnostics

Projecting the $M = 10\,000$ prior-simulation summaries to 2D via PCA and colouring by parameter value (Figure 2) reveals clear gradients for γ, moderate structure for β, and weak but non-trivial structure for ρ. This confirms the central identifiability challenge: β and ρ both suppress the epidemic — via higher transmission rate being countered by faster recovery (γ) or network rewiring (ρ) — so infection-curve summaries alone cannot fully separate them. Rewiring summaries carry independent information about both β and ρ.

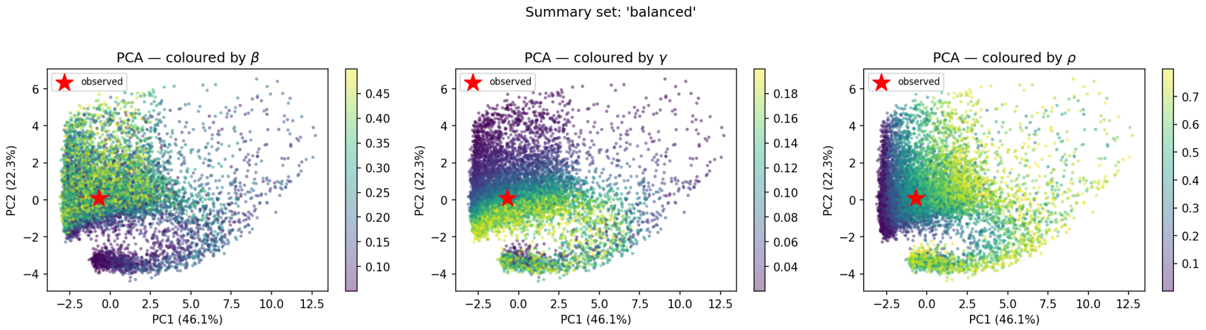

Figure 2: PCA diagnostic: prior simulations coloured by β, γ, ρ (left to right). Clear gradients for γ; partial confounding visible in β–ρ panels.

3.2 Summary-set comparison

We evaluated the single-source sets against the balanced set at $q = 1\%$ to understand which data streams carry information about which parameters. Table 1 and Figure 3 report posterior std relative to prior std (lower = more informative).

Table 1: Relative posterior std (std / prior std) per summary set at $q = 1\%$. Bold marks the best single-source set for each parameter.

Set	β	γ	ρ
infection_only	0.377	**0.084**	0.854
rewiring_only	**0.216**	0.804	**0.173**
degree_only	0.924	0.932	0.483
balanced (joint)	0.479	0.378	0.262

The single-source results reveal a clear separation of information: γ is almost entirely encoded in the infection curve (*infection_only*: 0.084), while β and ρ are best identified from rewiring summaries (*rewiring_only*: 0.22 and 0.17 respectively). The degree distribution alone is largely uninformative for β and γ but provides moderate constraint on ρ (0.48).

Crucially, no single-source set identifies all three parameters simultaneously. The *balanced* set is not the best for any individual parameter in isolation, but it is the only set that constrains all three to well below the prior width. This is precisely the effect of balanced modality representation: each data stream contributes to the joint distance rather than any one dominating it.

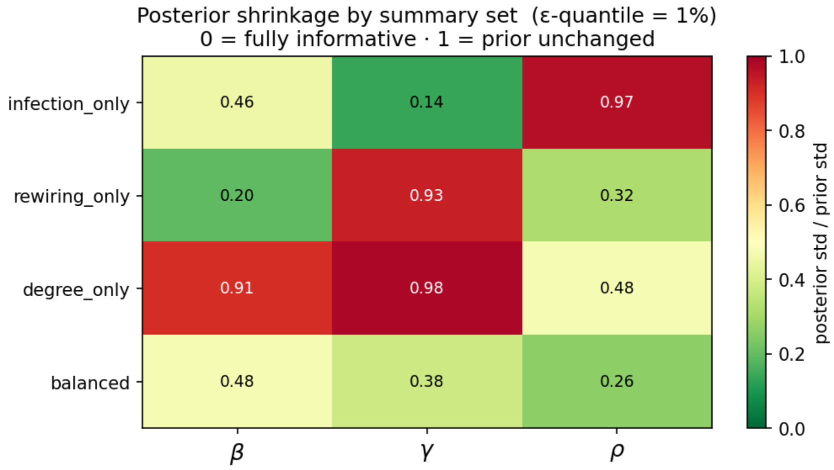

Figure 3: Posterior std (relative to prior) per summary set and parameter at $q = 1\%$. Lower is more informative.

4 Rejection ABC

We drew $M = 10\,000$ prior samples, computed the standardised Euclidean distance between each simulated and observed summary vector, and accepted draws below the 1% empirical quantile (Pritchard et al., 1999), giving $\varepsilon = 1.2852$ and $n_{\text{acc}} = 100$. The 1% quantile balances a variance–bias tradeoff: tighter tolerances reduce the approximation bias (accepted draws are closer to the true posterior) but yield fewer samples and a noisier posterior estimate, while looser tolerances give more stable estimates at the cost of a prior-biased posterior. This tradeoff motivates regression adjustment (Section 5.3), which corrects for the residual bias and allows a looser 5% tolerance. Figure 4 shows the marginal posteriors; Table 2 gives the numerical summaries.

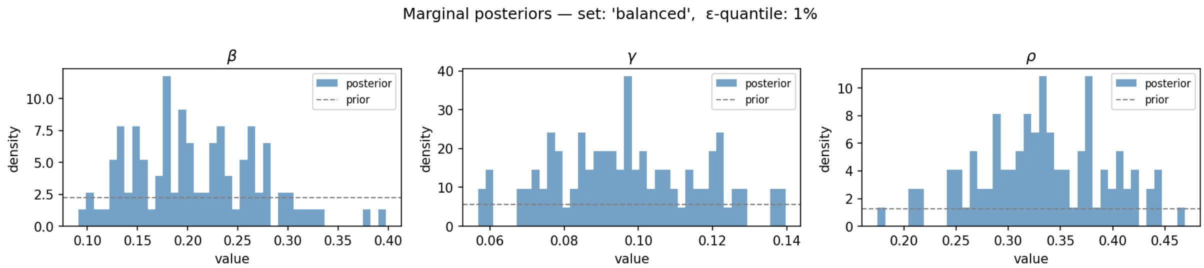

Figure 4: Rejection ABC marginal posteriors (blue) vs. uniform prior (dashed), *balanced* set, $\varepsilon = 1.2852$, $n_{\text{acc}} = 100$.

Table 2: Rejection ABC posterior ($q = 1\%$, *balanced*).

Param	Mean	Std	25%	Median	75%
β	0.208	0.063	0.159	0.201	0.254
γ	0.097	0.020	0.083	0.096	0.112
ρ	0.334	0.061	0.291	0.332	0.378

γ is best-identified (directly controls epidemic duration), β moderately so, and ρ is hardest to constrain — consistent with the PCA diagnostic and the negative (β, ρ) posterior correlation driven by their shared epidemic-suppression effect.

5 Advanced Methods

5.1 Semi-Automatic ABC (Fearnhead and Prangle, 2012)

Fearnhead and Prangle (2012) show that the optimal ABC summaries are the posterior means $\mathbb{E}[\theta \mid x]$, which can be approximated by fitting Ridge regression from the balanced features to each parameter on the pilot draws. In-sample R^2 values confirm strong predictability: $R^2_\beta = 0.812$, $R^2_\gamma = 0.857$, $R^2_\rho = 0.979$. Running rejection ABC in the resulting 3-dimensional projected space ($\varepsilon = 0.520$, $n_{\mathrm{acc}} = 100$) substantially tightens the posterior for β and provides moderate gains for γ and ρ (Figure 5, Table 3).

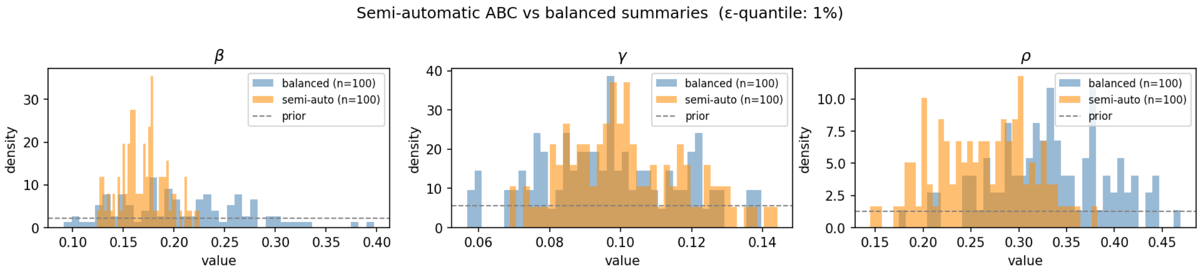

Figure 5: Balanced rejection ABC (blue) vs. semi-automatic ABC (orange), $q = 1\%$.

5.2 LASSO-Based Summary Selection (Blum et al., 2013)

Fitting LassoCV from summaries to each parameter on the pilot simulations retained 14/15 features for β ($\alpha = 6 \times 10^{-4}$) and all 15 for γ and ρ ($\alpha = 8 \times 10^{-4}$). No feature is entirely redundant, so the LASSO-selected set is effectively identical to the balanced set and yields the same posterior (Table 3).

Table 3: Posterior std and relative contraction (std/prior std) for rejection ABC variants at $q = 1\%$.

	β		γ		ρ	
Method	Std	Rel.	Std	Rel.	Std	Rel.
Prior	0.130	1.00	0.052	1.00	0.231	1.00
Balanced (15d)	0.062	0.48	0.020	0.38	0.061	0.26
LASSO (15d)	0.062	0.48	0.020	0.38	0.061	0.26
Semi-auto (3d)	0.023	0.18	0.017	0.33	0.051	0.22

5.3 Regression Adjustment (Beaumont et al., 2002)

After accepting draws at a loose tolerance ($q = 5\%$, $\varepsilon = 1.706$, $n_{\mathrm{acc}} = 500$), each accepted $\theta^{(i)}$ is corrected for the residual gap between its simulated summary $s^{(i)}$ and s_{obs}:

$$\tilde{\theta}^{(i)} = \theta^{(i)} - \hat{B}^\top (s^{(i)} - s_{\mathrm{obs}}),$$

where $\hat{B}$ is estimated by weighted least squares with Epanechnikov kernel weights (Beaumont et al., 2002). Figure 6 and Table 4 show that adjustment dramatically sharpens the posterior (σ_β: 0.099→0.024, σ_γ: 0.033→0.007, σ_ρ: 0.087→0.008). Crucially, the adjusted posterior is nearly tolerance-invariant for $q \lesssim 10\%$ (Figure 7), whereas the raw posterior degrades rapidly at loose tolerances.

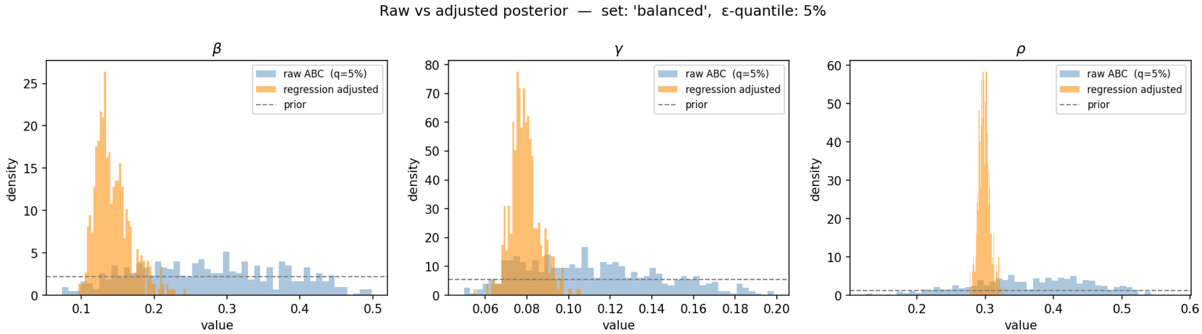

Figure 6: Raw ABC (blue) vs. regression-adjusted (orange) posteriors, $q = 5\%$, *balanced* set.

Table 4: Raw vs. regression-adjusted posteriors ($q = 5\%$).

Method	Param	Mean	Std	5%	Median	95%
Raw ($q = 5\%$)	β	0.276	0.099	0.127	0.272	0.439
	γ	0.111	0.033	0.066	0.108	0.169
	ρ	0.367	0.087	0.216	0.373	0.504
Adjusted	β	0.143	0.024	0.112	0.138	0.191
	γ	0.079	0.007	0.069	0.078	0.092
	ρ	0.299	0.008	0.286	0.298	0.313

5.4 Synthetic Likelihood (Wood, 2010)

At each MCMC proposal θ, we simulate $N_{\text{sim}} = 150$ datasets ($R_{\text{sim}} = 40$ replicates each), giving an N_{sim}/p ratio of 10.0 for our $p = 15$ summary features, and approximate the intractable likelihood as a multivariate Gaussian fitted to the N_{sim} summary vectors, using Ledoit–Wolf shrinkage for the covariance (Ledoit and Wolf, 2004):

$$\log \hat{p}_{\text{SL}}(s_{\text{obs}} \mid \theta) = \log \mathcal{N}(s_{\text{obs}};\, \hat{\mu}(\theta),\, \hat{\Sigma}(\theta)).$$

This is embedded in a Metropolis–Hastings MCMC with an **Adaptive Metropolis** (AM) proposal (Haario et al., 2001). Rather than fixing step sizes in advance, AM updates the proposal covariance online after an initial warm-up of $t_0 = 100$ steps:

$$\Sigma_{\text{prop}}(t) = \frac{2.38^2}{d}\,\text{Cov}(\theta_1, \ldots, \theta_t) + \epsilon I, \quad d = 3,$$

so the chain learns the posterior shape automatically and no manual step-size tuning is required (Roberts et al., 1997). Configuration: $N_{\text{steps}} = 2\,000$, burn-in $= 500$, initialised at the regression-adjusted posterior mean (0.143, 0.079, 0.299). The chain achieved an acceptance rate of 24.0%, consistent with the optimal AM target (Roberts et al., 1997).

Table 5: Synthetic likelihood MCMC posterior (1500 post-burn-in samples, thinning interval 31, 49 thinned draws; acceptance rate 24.0%).

Param	Mean	Std	5%	Median	95%	ESS
β	0.153	0.003	0.148	0.153	0.158	7.5
γ	0.081	0.002	0.078	0.081	0.084	11.9
ρ	0.301	0.005	0.295	0.300	0.310	10.2

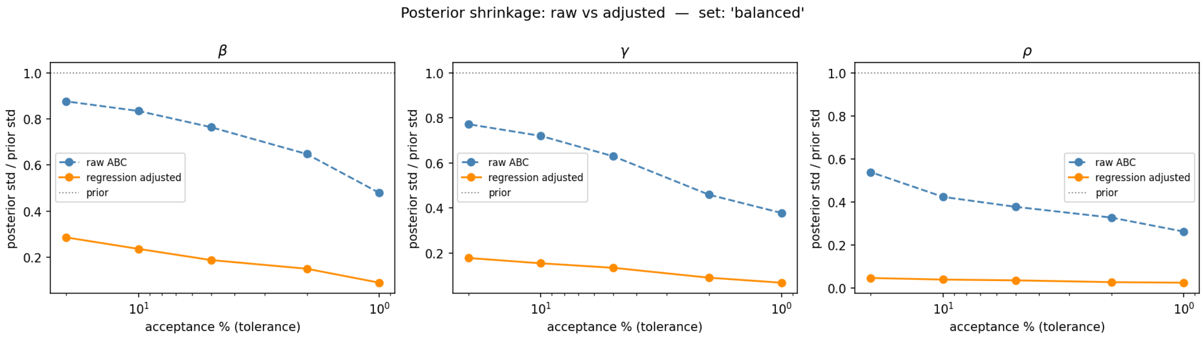

Figure 7: Posterior std (relative to prior) vs. tolerance q for raw (blue) and adjusted (orange) ABC. Adjustment is approximately tolerance-invariant for $q \lesssim 10\%$.

6 Comparison

Table 6 summarises all methods. Point estimates are consistent across advanced methods ($\hat{\beta} \approx 0.14$–0.15, $\hat{\gamma} \approx 0.079$–$0.081$, $\hat{\rho} \approx 0.299$–0.301). Regression adjustment gives the most reliable uncertainty quantification: large accepted set ($n = 500$), tolerance-robust, and verifiable via sensitivity analysis. The SL posterior is the sharpest, but comes with important caveats: the low ESS (7.5–11.9) means the posterior is estimated from fewer than 20 effectively independent draws despite 1500 post-burn-in samples, so the quoted standard deviations are unreliable. The tightness likely reflects the high $N_{\text{sim}} = 150$ per step combined with a multivariate normal approximation that may be overconfident for non-Gaussian summary statistics.

Table 6: Posterior standard deviations across all methods. ‡SL std is unreliable due to low ESS (7.5–11.9).

Method	σ_β	σ_γ	σ_ρ	Notes
Prior	0.130	0.052	0.231	–
Rej. ABC ($q = 1\%$)	0.062	0.020	0.061	Balanced, $n = 100$
LASSO ($q = 1\%$)	0.062	0.020	0.061	All 15 features kept
Semi-auto ($q = 1\%$)	0.023	0.017	0.051	3-d projection
Reg. adj. ($q = 5\%$)	0.024	0.007	0.008	$n = 500$, robust
Synth. lik.‡	0.003	0.002	0.005	$N_{\text{sim}} = 150$, accept. 24%

[TODO: If additional methods (ABC-MCMC, SMC-ABC) were implemented, add a subsection here.]

7 Conclusion

[TODO: Summarise: (1) identifiability findings; (2) how each method improves on basic rejection ABC; (3) SL mixing limitations and future work. Suggested length: ~0.5 pages.]

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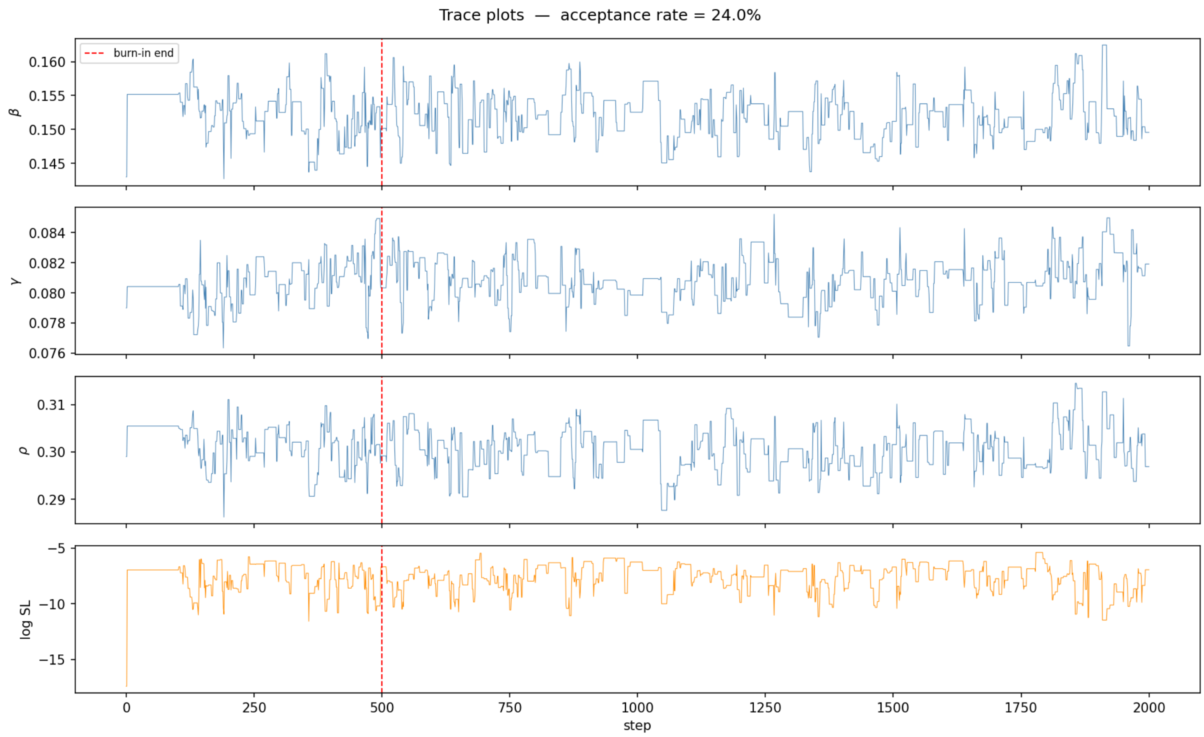

Figure 8: MCMC trace plots. The chain achieves an acceptance rate of 24.0% and mixes across the posterior, though the ACF decays slowly (thinning interval 31), resulting in ESS of 7.5–11.9.

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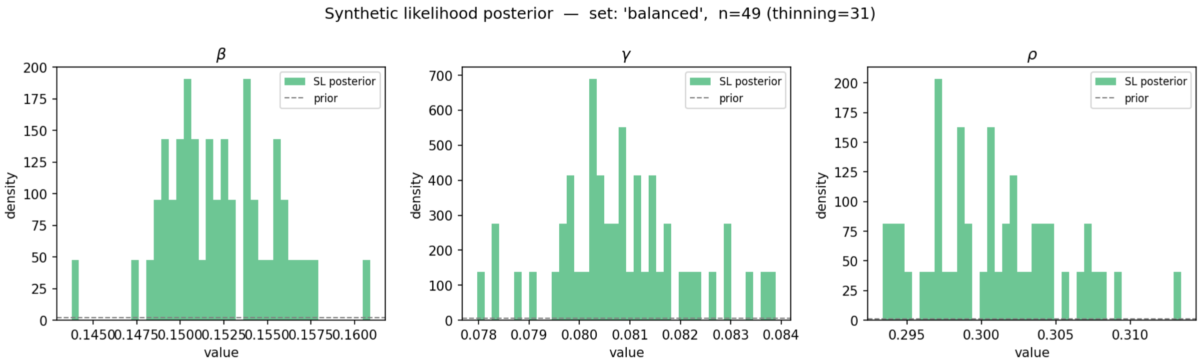

Figure 9: Synthetic likelihood MCMC posterior (green) vs. prior (dashed).